

**A SCALABLE FRAMEWORK FOR VISUALIZING SEGMENT-LEVEL  
PRECIPITATION USING GRIDDED WEATHER DATA**

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**ABSTRACT**

Weather substantially affects traffic operations, roadway safety, design, and overall transportation system performance. To support weather-related transportation research, multiple data collection methods have been developed; however, road sensor deployment remains costly and difficult to implement. While station-based weather data often lack spatial resolution, even gridded datasets with higher resolution typically do not align with roadway geometry, limiting their usefulness for road-specific weather analysis. To address this limitation, this study proposes a practical framework to map gridded precipitation data onto roadway segments to produce related metrics that are spatially consistent with the road network. This study conducted a pilot in Travis County, Texas, using one year of precipitation data to calculate segment-level metrics such as total precipitation, heavy rain days, etc. These metrics are then visualized on a user-friendly interactive platform designed to present result intuitively. To evaluate the generalizability of the proposed framework, the method was further applied to the Texas statewide network. The statewide application validated the scalability and adaptability of the framework across different regions and data sources. The results demonstrate its potential to support weather-related transportation studies such as planning, risk analysis, and operational decision-support.

*Keywords:* road weather, precipitation, gridded weather data, weather visualization

## 1 INTRODUCTION

2 Precipitation significantly affects transportation systems by impairing driver visibility, reducing  
3 pavement friction, and limiting vehicle maneuverability—consequently elevating crash risk, partic-  
4 ularly under heavy or sustained rainfall. More than 3,400 fatal crashes, over 191,000 estimated in-  
5 jury crashes and over 550,000 property-damage-only crashes occur in adverse atmospheric weather  
6 conditions in an average year (1). From 2019 to 2023, 12% of crashes in the United States were  
7 weather-related (1), with more than 73% linked to precipitation or mist. A study based on the Fa-  
8 tality Analysis Reporting System dataset from 1994–2012 shows 65% of adverse weather-related  
9 fatalities happened between November and April for 1994–2012 from the National Highway Traf-  
10 fic Safety Administration (2).

11 Besides the traffic safety, precipitation influences many other aspects of the transportation  
12 system, such as response time, congestion, road design, and mobility, etc. For instance, Zhan  
13 et al. found that increasing of 1 millimeter in hourly precipitation was associated with a 9.01  
14 second delay in median ambulance response time. Congestion may rise by up to 5.6% during  
15 rainfall events on workdays (4).. The precipitation is also one of the climatic variables required  
16 for predicting and assessing pavement performance to prevent premature cracking and rutting (5).  
17 Additionally, extreme rainfall caused flooding and road closures, disrupted mobility, and increased  
18 risks to residents' safety. Hurricane Harvey in 2017 resulted in over 80 fatalities (6), and more than  
19 100 people were killed in connection with the deadly Fourth of July flooding in 2025.

20 To support weather-resilient transportation planning and operations, considerable efforts  
21 have been made to collect and disseminate precipitation data. These efforts include ground obser-  
22 vations using rain gauges, disdrometers, ground radars, as well as the satellite estimates based on  
23 infrared and microwave signals (7). A most common used resource traditionally used by state de-  
24 partment of transportation (DOT)—such as those in Illinois (8), Minnesota (9), and Iowa (10)—is  
25 the fixed Road Weather Information System (RWIS) (9, 11), which consists mainly of Environ-  
26 mental Sensor Stations (ESS) along roadways to collect field data. Mobile RWIS technologies  
27 (12) have also been applied to fill the wide geospatial gaps in traditional fixed RWIS data.

28 Though RWIS is an effective tool to support road operations decision-making, the cost of  
29 properly installing is high, and the strategic plan is essential. The road weather information must  
30 be carefully examined, and the sensor types must be identified to generate the most desired data.  
31 Sensors not located on the roadway may be less accurate when predicting weather conditions on  
32 the roadway. Using them also requires data cleaning and harmonization across multiple sources  
33 with potentially varying reliability. Besides RWIS, precipitation data are collected and aggregated  
34 at regional, station, and gridded levels. The Texas Water Development Board (TWDB) maintains  
35 116 weather stations (13), and its interactive map viewer allows the public to access retail water  
36 service boundaries across Texas. To improve statewide coverage, TWDB has added other stations  
37 maintained by the National Weather Service and partnered with other regional and local entities,  
38 resulting in over 4,000 stations displayed in the TexMesonet system (14). This expansion provides  
39 more convenience and opportunities for data access. However, stations from different sources are  
40 designed based on different standards, such as varying update frequencies, which causes inconsis-  
41 tencies when trying to combine data to cover a larger area.

42 The Next Generation Weather Radar (NEXRAD) system (15), a network of 160 weather  
43 radars jointly operated by the National Weather Service, Federal Aviation Administration, and the  
44 U.S. Air Force, is another major source of precipitation data. The National Centers for Environ-  
45 mental Information (NCEI) provides access to both unprocessed base data and processed products

derived from radar via various algorithms. Gridded data from NEXRAD allow users to access a consistent source with better spatial resolution for large areas. The Multi-Radar Multi-Sensor (MRMS) system (16) integrates data from multiple NEXRAD radars and supplemental sources such as ground-based observations and satellites to generate high-resolution gridded meteorological products. Even though gridded data provide a high-resolution dataset, they still limit the resolution at which weather effects can be linked to individual roadway segments. This results in a significant spatial mismatch between observed weather patterns and the actual roadway environment where drivers operate.

Although many efforts have been made to collect and analyze precipitation data, challenges remain in presenting these data in an use-friendly format that is directly useful for transportation applications. While platforms like RITIS (17) support the visualization of roadway conditions including weather data, they include multiple datasets and are not specifically designed for weather-focused applications, making them difficult to replicate for similar use cases. And, many research projects use non-visualizable data formats or limit visualization to small areas, which further hinders timely replication and broader application by other practitioners. Additionally, due to the high cost of RWIS and the non-road-segment alignment of gridded data, practical applications remain limited.

To address the above limitations, a framework is proposed to generate precipitation estimates at the roadway segment level using publicly available gridded weather datasets, avoiding the need for deploying RWIS. As a demonstration, a pilot study was carried out in Travis County, Texas, incorporating one year of gridded precipitation data. These data were spatially aligned with roadway segments and visualized through an interactive platform. The resulting interface highlights key weather-related indicators—such as total precipitation, frequency of heavy rainfall, dry periods, and maximum hourly rainfall—to support localized assessments of weather exposure. This can assist roadway segment weather exposure, supporting research analysis and more discuss exploration among researchers, agencies and practitioners. To evaluate the broader applicability of the proposed framework, this study was also extended beyond the pilot area to cover the entire state of Texas using a different set of data.

In the rest of the paper, the data used in this study will be introduced first, followed by the methodology and then the result analysis. The paper concludes with a summary of findings and potential future directions.

## THE DATA

This section outlines the data sources employed in this study, classified into three categories: (1) meteorological data for obtaining precipitation measurements, (2) geometric data for roadway segment characteristics, and (3) Crash data for locating crash events near the corresponding roadway segments.

### Meteorological Data

To facilitate efficient spatial mapping to roadway segments, this project utilizes hourly gridded precipitation data—from July 2023 to June 2024—selected for its high spatial resolution and uniform coverage. The hourly data is integrated into this study to spatially map precipitation values onto corresponding roadway segments.

The primary data source is the NEXRAD system. NEXRAD detects precipitation and provides publicly available, gridded-based precipitation datasets at a spatial resolution of  $4 \text{ km} \times 4$



1 km ).

2 In addition to NEXRAD, this project incorporates data from the MRMS system. Of partic-  
3 ular relevance to this study is the quantitative precipitation estimation (QPE) dataset, which offers  
4 a spatial resolution of 1 km  $\times$  1 km and a temporal resolution as fine as 2 minutes. Hourly MRMS  
5 precipitation data is publicly available and can be accessed in GRIB2 format through the National  
6 Oceanic and Atmospheric Administration (NOAA) Big Data Program, which is hosted on Amazon  
7 Web Services (18).

8 To enhance accessibility, the project also leverages MRMS data made available by exter-  
9 nal institutions that offer user-friendly interfaces from the Iowa Environmental Mesonet (IEM),  
10 which provides MRMS precipitation data through a simplified web portal. The IEM platform (19)  
11 facilitates efficient data retrieval by timestamp and supports multiple temporal aggregation levels,  
12 including 2-minute, 1-hour, 24-hour, 48-hour, and 72-hour accumulations. This makes it especially  
13 suitable for rapid analysis and integration into transportation-related applications.

## 14 **Geometric Data**

15 The geometric dataset, comprising the geometries of the roadway segments and the boundaries of  
16 the study area, provides the spatial framework for the subsequent analysis.

### 17 *INRIX Segments*

18 In this project, INRIX traffic data(20) was utilized for the pilot study focused on Travis County,  
19 Texas. The INRIX reports traffic data for short directional freeway segments (typically under one  
20 mile), along with associated INRIX segments. INRIX segments represent discrete elements of the  
21 road network used to report traffic speeds. Each segment is classified into one of several functional  
22 road classes (FRC) to indicate its relative importance within the transportation network. This study  
23 focused on higher functional road classes by selecting highways and major intersections (FRC 1),  
24 major arterials (FRC 2), major roads (FRC 3). Lower-priority neighborhood streets (FRC 4 and  
25 higher) were excluded to ensure the analysis centered on the primary components of the regional  
26 traffic system. A total of 1,618 INRIX segments within Travis County were selected for analysis,  
27 comprising 110 segments classified as highways and major intersections, 594 as major arterials,  
28 and 914 as major roads. The average segment lengths for these three classes are approximately  
29 0.53 miles, 0.41 miles, and 0.33 miles, respectively, and a standard deviation is about 0.2 miles.

### 30 *Boundaries Geometry*

31 The boundary data is primarily used to allow users to define and select interest areas for visualiza-  
32 tion purposes. In this project, boundaries polylines for both Texas Department of Transportation  
33 (TxDOT) districts(21) and Texas counties(22) are incorporated to support regional analysis and  
34 filtering. Texas is divided into 25 districts and 254 counties, and both boundary layers are utilized  
35 to enable flexible geographic selection. For visualization, weather data is displayed only for the  
36 districts or counties selected by the user. This approach ensures clarity and relevance by limiting  
37 the presentation to areas directly involved in the analysis.

## 38 **Crash Data**

39 To supplement the roadway segment information, this project also incorporates crash data to an-  
40 alyze incidents occurring along the roadway network. In Texas, crash data is available through  
41 the Crash Records Information System (CRIS)(23), maintained by the TxDOT. Publicly accessible

1 data can be obtained through the CRIS Query Tool, which enables users to download and analyze  
 2 customized crash records. Consistent with the meteorological data collection period, crash data  
 3 from July 2023 to June 2024 was obtained for this study. During this period, a total of 17,858  
 4 crashes were recorded in Travis County.

## 5 **METHODOLOGY**

6 This section outlines the implementation of the roadway visualization web interface, detailing the  
 7 procedures for raw data processing and web interface layout.

### 8 **Data Preparation Procedures**

9 Meteorological and crash records are spatially mapped to corresponding counties and roadway  
 10 segments to support location-specific analyses.

#### 11 *Meteorological data Mapping*

12 Each roadway segment in Travis County pilot study is identified by a unique INRIX segment ID  $s_i$ ,  
 13 where  $i = 1, 2, \dots, 1,618$  indexes the 1,618 selected segments. A total of 14,212,512 precipitation  
 14 records were generated by mapping hourly NEXRAD grid precipitation data across 1,618 roadway  
 15 segments over 366 days (July 2023 to June 2024). The geographic location of each segment is  
 16 represented by the coordinates of its starting point longitude  $x_i^{irx}$  and latitude  $y_i^{irx}$ . Each hourly  
 17 timestamp  $t$  corresponds to one of the hourly intervals during the one-year period. To assign  
 18 precipitation values, each segment's start point was matched to the nearest static NEXRAD grid  
 19 cell, represented as longitude  $x_i^{nxd}$  and latitude  $y_i^{nxd}$ . For each timestamp  $t$ , the corresponding  
 20 precipitation value  $p_{i,t}$  was extracted from the NEXRAD grid data. The mapped precipitation  
 21 sample data is exhibited in Table 1a, where all precipitation values are expressed in inches. The  
 22 average distance between INRIX segment start points and their nearest NEXRAD grid cell centers  
 23 for Travis County is 1.42 km, with a minimum distance of 0.04 km and a maximum of 2.70 km.  
 24 The extracted segment-hour precipitation values  $p_{i,t}$  were subsequently used to compute weather-  
 25 related variables for each segment-hour observation.

#### 26 *Crash data Mapping*

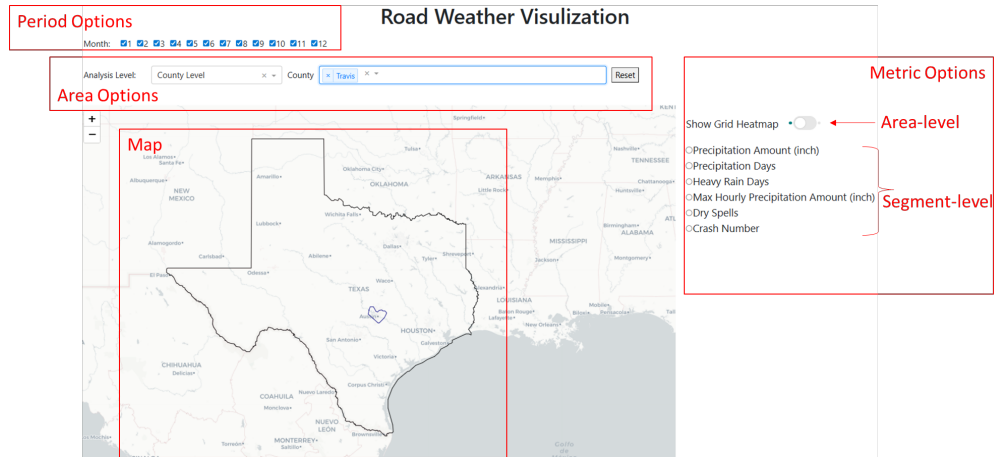
27 Crash reports without valid geographic coordinates were excluded from the analysis, as they could  
 28 not be reliably associated with the roadway network. After filtering, a total of 16,852 crash in-  
 29 cidents remained in Travis County. Each crash location was then spatially matched to its nearest  
 30 highway segment using a nearest-neighbor approach. To ensure reliable association with the road-  
 31 way network and segment-level weather conditions, only crashes within a 30-meter radius of a  
 32 roadway segment were retained. Consequently, 7,080 crash incidents were successfully mapped to  
 33 the selected INRIX segments in Travis County. Formally, each crash  $c_n$  occurred on date  $d_n$ , where  
 34  $n \in \{1, \dots, 7,080\}$ , with coordinates  $(x_n^{cris}, y_n^{cris})$ , was assigned to the nearest INRIX segment  $s_n$ ,  
 35 where  $s_n \in \{s_i\}$  denotes the identifier of the polyline segment. The spatial association was deter-  
 36 mined by computing the shortest distance between the crash location and the polyline geometry,  
 37 selecting the closest segment within the 30-meter. Table 1b presents a sample of the mapped crash  
 38 data.

**TABLE 1:** Sample of Processed Data

| (a) Sample of Processed Precipitation Table |           |                     |           | (b) Sample of Processed Crash Reports Table |          |            |            |
|---------------------------------------------|-----------|---------------------|-----------|---------------------------------------------|----------|------------|------------|
| $i$                                         | $s_i$     | $t$                 | $p_{i,t}$ | $n$                                         | $c_n$    | $d_n$      | $s_i$      |
| 1                                           | 166980906 | 2023-07-01 00:00:00 | 0.0       | 1                                           | 19633973 | 2023-07-01 | 1595314919 |
|                                             | ...       |                     |           | 2                                           | 19645837 | 2023-07-01 | 1595401121 |
| 14210653                                    | 14210652  | 2024-06-20 22:00:00 | 0.0197    |                                             | ...      |            |            |
| 14210654                                    | 14210652  | 2024-06-20 23:00:00 | 0.0050    | 7079                                        | 20263858 | 2024-06-30 | 1595163018 |
|                                             | ...       |                     |           | 7080                                        | 20263861 | 2024-06-30 | 1595281953 |

## 1 Interactive Interfaces Design

2 Figure 1 illustrates the layout of the designed user-interactive interface. The period option section  
 3 allows for multiple selections in a 12-month range. Below it, the area section enables users to  
 4 choose between district-level or county-level options, as well as specific districts and counties for  
 5 the chosen level. On the right side, the metric section lists all variables described aforementioned.  
 6 The central part of the interface features a map, which supports zooming in and out to visualize  
 7 the selected segment-level metrics.

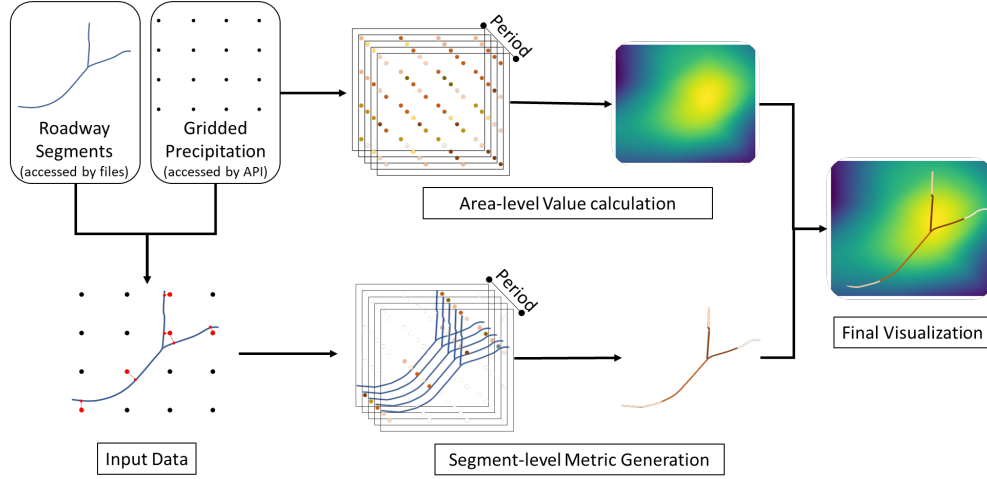
**FIGURE 1:** Layout of the Interface

## 8 METRIC DEFINITIONS

9 In the user interactive interface, two types of precipitation visualization are supported: (1) area-  
 10 level heatmaps and (2) roadway segment-level heatmaps. The workflow of producing these two  
 11 level aggregate metrics is illustrated in Figure 2.

12 To support user-defined monthly selection, the area-level precipitation data are pre-aggregated  
 13 and stored by month for efficient rendering. While roadway segment-level visualization retains  
 14 daily resolution to support the computation of day-specific variables, such as the precipitation  
 15 days, even though the interface displays results by month. Hourly data are therefore aggregated to  
 16 the appropriate temporal resolution — monthly for area-level variables and daily for segment-level  
 17 variables — to balance computational accuracy and visualization efficiency.

18 For the area-level precipitation heatmap, the monthly accumulated precipitation for each  
 19 MRMS grid cell is denoted as  $p_{g,m}$ . It is computed by aggregating the hourly precipitation values

**FIGURE 2:** Workflow of Road Segments Precipitation and Visualization

- 1  $p_{g,t}$  for grid cell  $g$  over all time intervals  $\tau$  within the selected month  $m$ , as shown in Equation (1).
- 2 The intervals  $\tau$  here is 72 hours interval downloaded from IEM.

$$3 \quad p_{g,m} = \sum_{\tau \in m} p_{g,i\tau} \quad (1)$$

4 While area-level precipitation heatmaps are included, this study primarily focuses on road-  
 5 way segment-level visualization. The following section details the derivation and daily-level ag-  
 6 gregation of the associated variables.

### 7 *Total Precipitation*

8 Total precipitation is defined as the cumulative precipitation over a given period on a roadway  
 9 segment. The daily precipitation amount for segment  $s_i$  on day  $d$ , denoted as  $p_{i,d}$ , is calculated by  
 10 summing the corresponding hourly precipitation values  $p_{i,t}$ , where  $t$  is the hourly timestamp over  
 11 July 2023 to June 2024, as shown in Equation 2:

$$12 \quad p_{i,d} = \sum_{t \in d} p_{i,t} \quad (2)$$

### 13 *Precipitation Days*

14 In addition to reporting daily precipitation amounts, a binary variable  $\delta_{i,d}$  is defined to indicate  
 15 whether a given day  $d$  on roadway segment  $s_i$  is classified as dry. A day is considered dry if the  
 16 total daily precipitation  $p_{i,d}$  is less than a predefined threshold  $\theta_p$ , which is set as 0.01 inch (0.254  
 17 mm)(24) in this study. The definition of  $\delta_{i,d}$  is given in Equation 3:

$$18 \quad \delta_{i,d} = \begin{cases} 1, & \text{if } p_{i,d} < \theta_p \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

19 Then the total number of precipitation days for segment  $s_i$  in month  $m$ , denoted as  $p_{i,m}^{dry}$ , can  
 20 be computed by counting the number of non-dry days. Based on the binary indicator  $\delta_{i,d}$ ,  $p_{i,m}^{dry}$  can  
 21 be expressed as:

$$p_{i,m}^{dry} = \sum_{d \in m} (1 - \delta_{i,d}) \quad (4)$$

## Heavy Rain Days

To identify days characterized by heavy rainfall, a rule-based approach was applied using hourly precipitation data  $p_{i,t}$ . A day is classified as a heavy rain day if any 6 consecutive hours within that day meet both of the following criteria (25): (1) measurable precipitation, defined as at least  $\theta_h = 0.01$  inches, is recorded in each hour; and (2) the total accumulated precipitation during these six hours exceeds the threshold  $\Theta_h = 0.1969$  inches (i.e., 5 mm). To allow for brief break in rainfall (26, 27), the method also accepts a 7-hour window in which at least 6 of the 7 hours record measurable precipitation. The single non-precipitating hour is permitted only if it is immediately preceded and followed by hours with measurable precipitation. The total precipitation during this 7-hour window must still exceed  $\Theta_h$ . If these criteria is met within day  $d$  on segment  $s_i$ , a binary indicator  $p_{i,d}^h$  for heavy rain is assigned a value of 1; otherwise, it is set to 0.

$$p_{i,d}^h = \begin{cases} 1, & \text{if heavy rain condition is met on day } d \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

## Maximum Hourly Precipitation

Intense rainfall events can cause significant disruptions and hazards, especially when they occur suddenly, leaving limited time for residents to respond. To quantify such extreme precipitation, the maximum hourly precipitation during a period is pointed out as a metric as well. The variable  $p_{i,d}^{max}$  represents the maximum hourly precipitation on day  $d$  for roadway segment  $s_i$ , as defined in Equation (6). The maximum value of  $p_{i,d}^{max}$  within a given month  $m$  is used to represent the peak hourly precipitation in this month for that segment.

$$p_{i,d}^{max} = \max\{p_{i,t} \mid t \in d\} \quad (6)$$

## Dry Spells

The variable  $p_{i,d}^y$  represents the length of the dry spell to which day  $d$  belongs for segment  $s_i$ . A dry spell is defined as a maximal sequence of consecutive days with no measurable precipitation, i.e., days for which  $\delta_{i,d} = 1$  (28). For each dry day,  $p_{i,d}^y$  is assigned the total number of consecutive dry days in that spell. For non-dry days ( $\delta_{i,d} = 0$ ),  $p_{i,d}^y$  is set to 0.  $\text{spell}(i,d)$  denotes the maximal sequence of consecutive dry days ending on day  $d$  for segment  $s_i$ . To capture the potential impact of long dry periods, only dry spells lasting at least five consecutive days are considered indicative of significant dry conditions (28). For each segment  $s_i$  and month  $m$ , the variable  $p_{i,m}^y$  denotes the number of distinct dry spells in that month that meet this threshold.

## Crash Number

To quantify the daily number of crash incidents on each roadway segment, the variable  $p_{i,d}^c$  denotes the total number of crashes mapped to INRIX segment  $s_i$  on day  $d$ . The mapping procedure is detailed in Section “Crash data Mapping”. Let each crash be indexed by  $n$ , where  $n \in \{1, \dots, 7,080\}$  corresponds to the subset of crashes successfully linked to the selected INRIX segments. For a given day  $d$  and segment  $s_i$ , all crashes  $n$  satisfying  $d_n = d$  and  $s_n = s_i$  are counted, as shown in

1 Equation (7).

$$2 \quad p_{i,d}^c = \sum_{\substack{d_n=d \\ s_n=s_i}} 1 \quad (7)$$

### 3 RESULTS AND VISUALIZATION

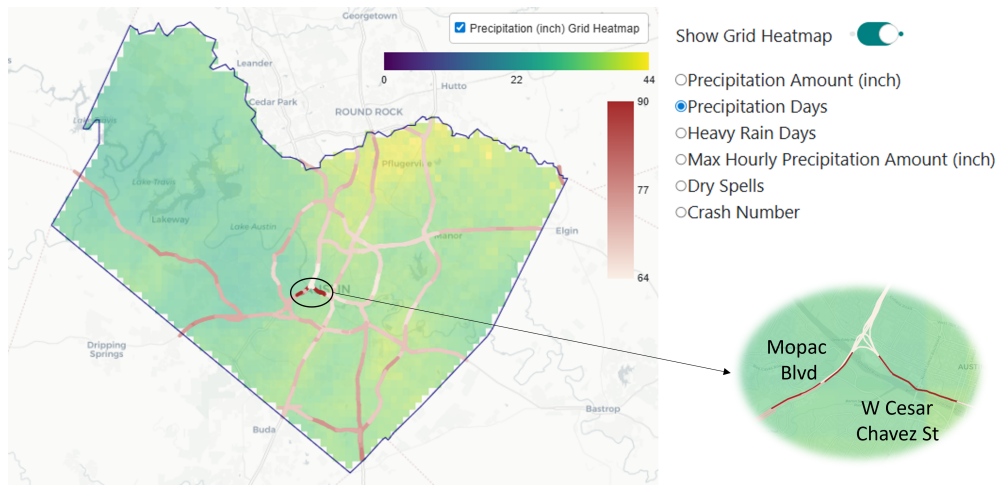
4 The background of area-level heatmap on Figures 3,4,5, 6, and 9 presents the spatial distribution  
5 of total precipitation in Travis County, aggregated across all grid cells from July 2023 to June  
6 2024. Overall, precipitation is concentrated in the eastern part of the county, particularly in the  
7 north-central area.

#### 8 Segment-Level Precipitation Characteristics

##### 9 *Number of Precipitation Days*

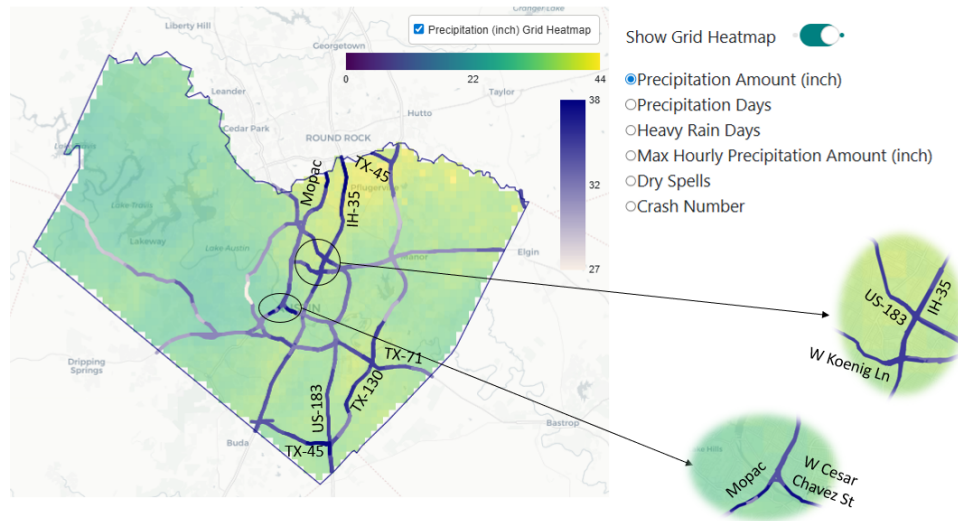
10 Figure 3 shows the total number of precipitation days for each roadway segment in Travis County.  
11 Segments located near the Colorado River, especially around the intersection of Mopac Blvd and  
12 W Cesar Chavez St, experienced the highest number of precipitation days (around 90 days). In  
13 contrast, most other segments had a relatively consistent number of precipitation days, with 75%  
14 of the segments experiencing approximately 70 days of precipitation.

**FIGURE 3:** The precipitation days on roadway segments in Travis County



##### 15 *Total Precipitation Amount*

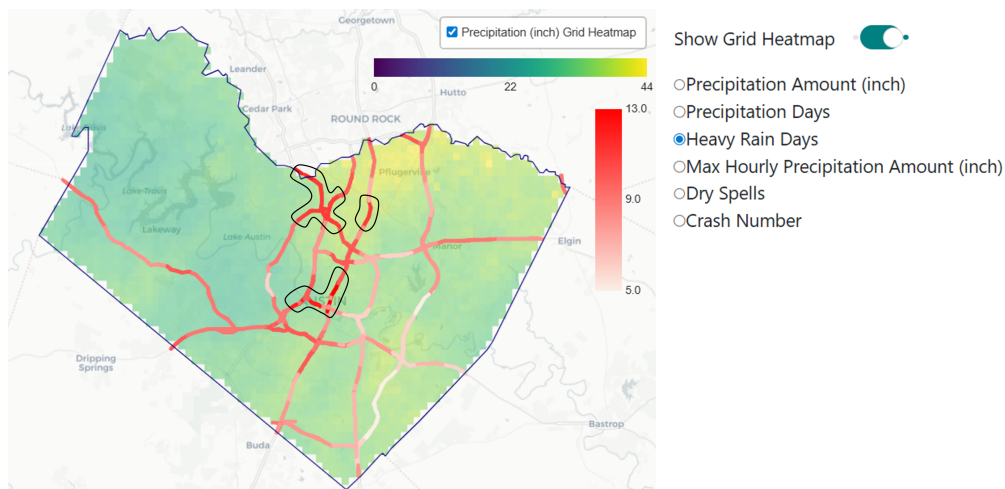
16 The total annual precipitation on roadway segments generally aligns with the area-based heatmap.  
17 Segments in the north-central and southeastern parts of the Travis County received more precipi-  
18 tation than others. Specifically, several segments recorded more than 35 inches of annual rainfall,  
19 including: TX-45-TOLL (both northern and southeastern sections), TX-130-TOLL (southeast),  
20 TX-71 W near its intersection with TX-130, US-183 (north and southeast), IH-35 (north), Mopac  
21 Blvd (north), the Mopac/W Cesar Chavez St area near the Colorado River, and the central area  
22 encompassing US-183, I-35 S, and W Koenig Ln.

**FIGURE 4:** The annual precipitation amount on roadway segments in Travis County

### 1 Heavy Rainfall Days

2 The spatial distribution of heavy rain days differs from the total precipitation amount. For example,  
 3 while TX-45-TOLL in southeast Travis County received high total rainfall, it did not experience a  
 4 particularly high number of heavy rainfall days. Only a few segments—those in the northern part  
 5 of the County, IH-35 north, and the Mopac Blvd/W Cesar Chavez St area—experienced more than  
 6 10 heavy rainfall days during the study period.

7 At the county level, 18.7% of the days with any recorded precipitation involved heavy  
 8 rainfall. When considering all segment-day combinations with rain ( $\delta_{i,d} = 1$ ), 12% of those cases  
 9 were classified as heavy rain events. The difference reflects that while heavy rain may occur on a  
 10 given day, it does not necessarily affect all segments uniformly.

**FIGURE 5:** The total heavy rain days on roadway segments in Travis County

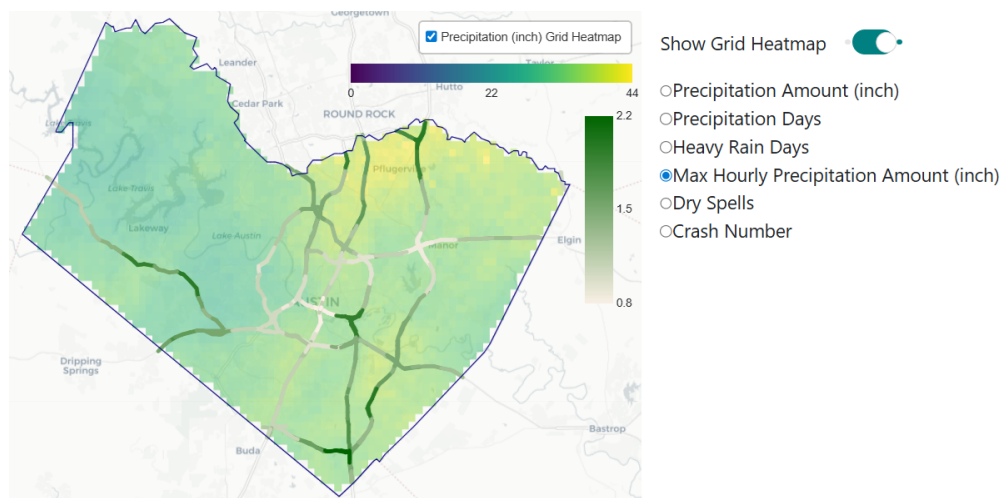


### 1 *Maximum Hourly Precipitation*

2 The maximum hourly precipitation amount among July 2023 and June 2024 was illustrated in  
3 Figure 6. The map shows that segments in surrounding may experienced the sudden heavy rain  
4 rather than segments in middle area. Though most of these segments didn't experience lot heavy  
5 rain days and large amount, the heavy hourly rain is still needs attention....

6 Figure 6 illustrates the maximum hourly precipitation recorded on each roadway segment  
7 between July 2023 and June 2024. Notably, several segments in the edge areas of the county  
8 experienced brief but intense rainfall events. Although these segments may not have recorded high  
9 total precipitation or frequent heavy rain days, sudden extreme hourly rainfall remains a potential  
10 hazard requiring attention.

**FIGURE 6:** The maximum hourly precipitation amount on roadway segments in Travis County



### 11 *Dry Spells*

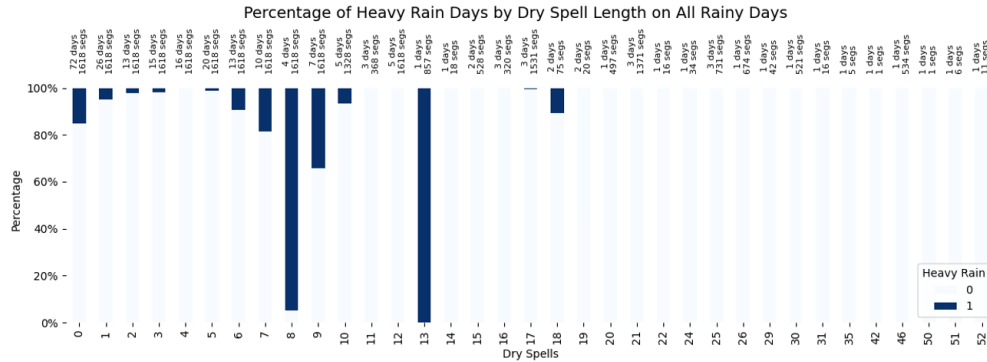
12 Dry spells across segments ranged from 1 to 52 consecutive days, with a mean duration of 7 days.  
13 Most segments experienced approximately 20 dry spells lasting longer than 5 days. Segments in  
14 central Travis County tended to have more such dry spells (up to 22), while those in the southern  
15 area had slightly fewer (around 18 days).

16 As shown in Figure 7, the majority of rainy days followed relatively short or no dry spells.  
17 As the dry spell length increases, the number of observations decreases substantially. Among  
18 rainy days following short dry spells (fewer than 5 days), the proportion of heavy rainfall events  
19 remained relatively low. However, when dry spells lasted between 7 and 10 days, the likelihood of  
20 heavy rainfall increased noticeably. Due to the small number of observations, cases with dry spells  
21 longer than 10 days are not considered here.

### 22 *Crash Frequency and Rain Conditions*

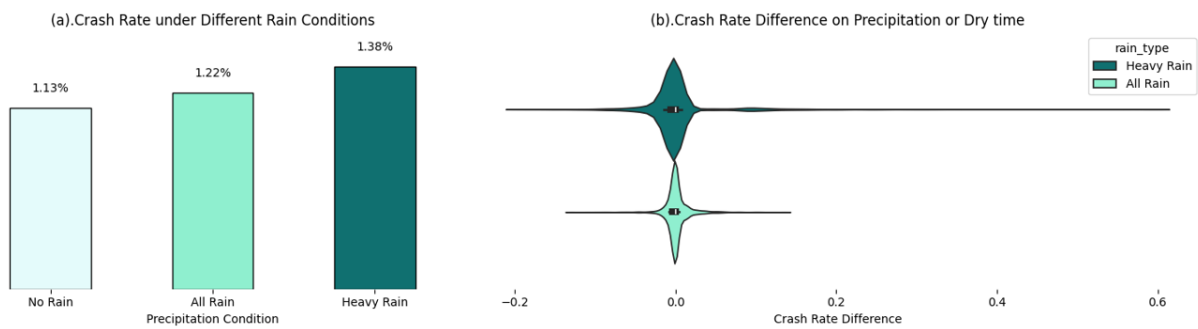
23 The annual crash number per segment ranged from 0 to 148 over the study period. Most crashes  
24 occurred along IH-35 corridor, as well as on Mopac Blvd and US-183 in the northern part of the  
25 County. At the segment-day level, crashes were rare: 98.86% of observations ( $p_{i,d}^c$ ) reported no  
26 crashes. Figure 8(a) shows the crash rate was 1.13% on non-rainy days, 1.20% on rainy days,  
27 and 1.35% on heavy rain days. This indicates a slight increase in crash likelihood during rainfall,



**FIGURE 7:** The percentage of heavy rain days by dry spells on all rainy days

1 especially under heavy rain. Figure 8(b) further compares the difference in crash rates between  
 2 rainy and non-rainy conditions across segments. In both cases, there is no significant difference  
 3 on crash happened on rain day or dry day, neither the heavy rain day and non heavy rainy day.  
 4 However, for heavy rain conditions, the distribution exhibits a longer positive tail, suggesting that  
 5 for a small subset of segments, crash rates under heavy rain were up to 60% higher compared to  
 6 non-heavy rain conditions.

7 To further examine the relationship between rainfall and crash occurrence, a logistic re-  
 8 gression was conducted using a binary dependent variable (i.e., if  $p_{i,d}^c$  is large than 0) indicating  
 9 whether a crash occurred on a given segment-day. The logistic regression results show a coeffi-  
 10 cient of 0.0643 ( $p = 0.046$ ) for rainy days and 0.1446 ( $p = 0.068$ ) for heavy rain days. Although  
 11 both coefficients indicate a positive relationship with crash occurrence, only the rainy day effect is  
 12 statistically significant at the 5% level. The heavy rain effect, despite being larger in magnitude,  
 13 does not meet the conventional threshold for significance.

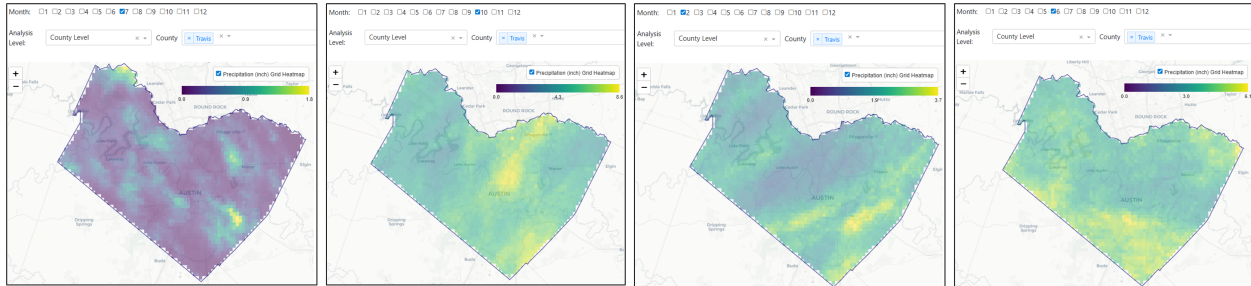
**FIGURE 8:** Crash rate on different precipitation conditions

## Months

Users can explore precipitation patterns on the interactive web application by selecting only the months of interest. Figure 9 includes examples from July and October 2023, and February and June 2024. July exhibited minimal precipitation, with only a few localized spots receiving small amounts. October brought more substantial rain, especially along the IH-35 corridor north of

downtown. In February, precipitation mostly occurred in the southeastern part of the county, whereas in June, the rainfall was concentrated more in the southern region. These examples reveal how precipitation patterns can change both spatially and temporally. Understanding these monthly changes can be useful for scheduling roadway activities that depend on weather.

**FIGURE 9:** Precipitation heatmap examples by month



It is important to note that this pilot study is based on data from only a single year—from July 2023 to June 2024—which is insufficient for drawing conclusions about monthly precipitation trends. For example, in July 2023, less than 4 days in Travis County were recorded as having a very minimum precipitation. In contrast, July 2025 presents a markedly different pattern, with additional results discussed in the following section. To identify more reliable seasonal precipitation trends, a multi-year dataset is required.

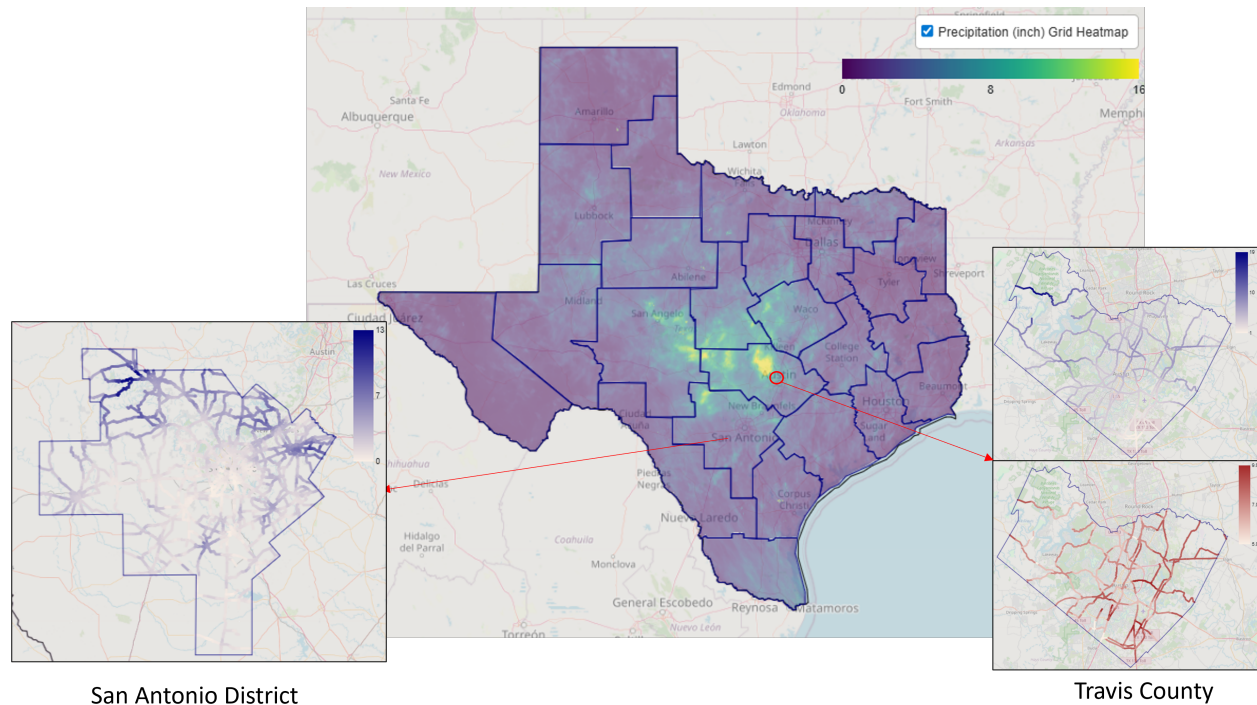
### Generalization to Statewide Scale

The proposed roadway visualization approach is designed to support generalizable applications. To demonstrate its scalability, an extended application at the statewide level was conducted, encompassing precipitation data over the 10-day period from July 1 to July 10, 2025.

For this statewide analysis, the TxDOT Roadway Inventory – On-System dataset (29) was used to represent the highway network maintained by the state. The on-system network includes 167,495 segments, with an average length of 0.44 miles and a standard deviation of 0.85 miles. To support the statewide analysis, MRMS grid data were retrieved and incorporated both as background heatmaps and as segment-level precipitation variables. Figure 10 presents the cumulative grid precipitation across Texas during the selected 10-day period. As shown, central Texas experienced the highest precipitation totals, with localized accumulations reaching up to 16 inches. Segment-level precipitation data is viewed by either district or county.

The total precipitation amount in the San Antonio District is highlighted in the lower-left of Figure 10. Notably, Kerr County—located in the northwest part of the district—experienced severe precipitation, which resulted in the widely reported July 4 flood. Segments along I-39 and its spur, FM 1340, recorded peak hourly precipitation rates of approximately 3 inches when the Max Hourly Precipitation variable is visualized. In addition to Kerr County, Guadalupe County also recorded peak hourly rates exceeding 3 inches, although its total accumulated precipitation was comparatively lower.

The lower-right of Figure 10 shows segment-level precipitation in Travis County. In contrast to the pilot study period in July 2023—which saw only a few days of light rainfall throughout the month—many segments in Travis County experienced between 5 and 10 days of measurable

**FIGURE 10:** The precipitation heatmap of all district in Texas

1 precipitation within this 10-day period. While the eastern portion of the county had more fre-  
 2 quent precipitation days, the highest total rainfall and peak hourly intensities were observed along  
 3 segments of East Farm to Market 1431 Road in the western part of the county.

#### 4 CONCLUSION

5 This study explored a practical method to map public meteorological datasets to roadway seg-  
 6 ments, and then generate the weather variables based on the mapped precipitation value. The pilot  
 7 study, first performed for Travis County, Texas, provided insights into segment-level precipitation  
 8 variability, dry periods, and high-intensity events. The segment-crash associations, while modest  
 9 overall, pointed to specific corridors where rainfall may coincide with increased crash rates and  
 10 as an example to show the potential to add other segment events that users are interested in. These  
 11 findings show how segment-level weather information can help in understanding local risks, even  
 12 though they are based on just one year of data. The results could be further improved by including  
 13 more years of observations. After the pilot study, the method was tested across the Texas highway  
 14 system during a 10-day time window in July 2025. This test showed that the approach can be  
 15 conducted promptly even when working with different road networks and precipitation data. It  
 16 suggests the method can be replicated elsewhere without needing much adjustments, as long as  
 17 the necessary road layout and gridded rainfall information are available. A key benefit lies in its  
 18 reliance on publicly nationwide available inputs, making it suitable for broader use without requir-  
 19 ing costly sensor infrastructure and adjust different source data to a same standard. The interactive  
 20 web tool improves user access by letting them view weather exposure patterns geographically.

21 Though the method is easy to replicate and relies on publicly available data sources, several

foreseeable challenges may arise. One challenge is the selection of appropriate roadway segment geometry. In this study, the TxDOT Roadway Inventory – On-System dataset was selected for the statewide application. However, the average segment length in this dataset is 0.44 miles but a standard deviation is 0.85 miles, which means there are many very short segments. These short segments significantly increase processing workload, as each segment requires individual precipitation value mapping, and actually such small granularity may not be necessary for weather related project. Selecting an appropriate segmentation dataset is therefore crucial for optimizing the framework's efficiency. Another challenge would be the substantial volume of gridded precipitation data required. Since hourly precipitation data is used, the data volume grows rapidly, especially if multiple years are included in the future analysis. The mapping process also requires substantial computational effort when the project is extended. For instance, in the statewide application, 167,495 segments must be matched with hourly precipitation data for all relevant years. These demands highlight the importance of optimizing the framework when extending the proposed framework to larger regions and longer timeframes.

Future work could continue the current framework by adding more weather variables - such as temperature, wind - to provide a more comprehensive representation of weather information. Update the framework visualization by including multiple years of data would support a more reliable assessment of long-term impact trends. Linking segment-level weather information with other indicators such as crash records, speed profiles, or capacity, outputs could enhance the understanding the mutual influence. Advancing interactive and user-friendly visualization tools as part of a broader effort to develop platforms in an user friendly format would further improve the practical utility of the data for planners, researchers, and operators. Potential use cases could be modifying traffic signal timings considering the adverse weather condition, improving corridor travel time estimation by incorporating the precipitation value, implementing dynamic variable speed limits as needed, and supporting transportation management systems such as the real-time warning system, among others. These application examples reflect the broader value of segment-level weather data in supporting safer and more efficient transportation systems.

## **AUTHOR CONTRIBUTIONS**

The authors confirm their contributions to the paper as follows: study conception and design: N. Ruiz-Juri and Y. Li; data acquisition: A. Kosicki and S. Hossain and Y. Li; data analysis and interpretation: Y. Li; visualization: Y. Li; manuscript drafting: Y. Li, A. Kosicki, N. Ruiz-Juri. All authors reviewed the results and approved the final version of the manuscript.

## **DECLARATION OF CONFLICTING INTERESTS**

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